YONG JIN (JINA) CHUN

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SUMMARY

Yong Jin (Jina) Chun is a PhD student in Software Engineering at the University of California, Irvine, advised by Professor Iftekhar Ahmed. She completed a Master's degree in Software Engineering from the University of California, Irvine. Her research focuses on Agentic AI and Large Language Models (LLMs), with an emphasis on how autonomous agents collaborate, communicate, and reason together to solve multi-step tasks. She designs and evaluates multi-agent frameworks that improve the reliability, reasoning, and scalability of AI systems, with applications in automated software development. She also serves as a reviewer for top-tier journals such as IEEE Transactions on Reliability, ACM Transactions on Software Engineering and Methodology (TOSEM), and Empirical Software Engineering (EMSE).

EDUCATION

PhD in Software Engineering, University of California, Irvine — *Sept 2022 - Current*
Advisor: Professor Iftekhar Ahmed — GPA: 4.0/4.0

MS in Software Engineering, University of California, Irvine — *Sept 2022 - Sept 2025*
Advisor: Professor Iftekhar Ahmed — GPA: 4.0/4.0
Dissertation: Multi-Agent Debate for Advancing Agentic Software Development

BS in Computer Science , Boston University — *Aug 2017 - May 2021*
Cum Laude, Dean's Honor List — GPA: 3.6/4.0

WORK EXPERIENCES

Research Intern, — *June 2026 – Sept 2026*
SAINTES Group, Advisor: Christian Bird — Redmond, Washington

- Designed a working-knowledge memory system that integrates across agent sessions to enable contextual retrieval throughout the development lifecycle.
- Explored indexing, retrieval, and privacy-aware redaction strategies to strengthen agent-assisted software engineering workflows.

Research Assistant, — *Sept 2022 – Current*
STAIRS lab — University of California, Irvine

- Published two peer-reviewed papers on Large Language Models (LLMs) in software engineering.
- Developed multi-agent debate frameworks to analyze limitations and enhance the performance of LLM-driven software engineering tasks.
- Conducted an empirical analysis of prompting techniques across diverse software engineering tasks to investigate factors influencing their effectiveness.

Data Science Intern, — *Jan 2020 – Apr 2020*
Orion Health — Auckland, New Zealand

- Analyzed data annotation strategies to enhance accuracy and efficiency in clinical data processing.

- Developed a Natural Language Processing (NLP)-based annotation tool for clinical data.

Software Engineering Intern, *Jun 2019 – Aug 2019*
Parasoft Southeast Asia Singapore

- Investigated the company's automated software testing product to identify software quality issues.
- Designed and developed a user interface improvement based on the findings.

PUBLICATIONS

Jina Chun, Iftekhar Ahmed (2026). What Do Agents Communicate? Characterizing Information Exchange in Multi-Agent Systems. Manuscript submitted for publication in the *Conference on Neural Information Processing Systems (NeurIPS).*

Jina Chun, Qihong Chen, Jiawei Li, Iftekhar Ahmed (2026). Enhancing LLM Performance Through Debate: An Empirical Study on Multi-Agent Debate for Software Engineering Tasks. *ACM Transactions on Software Engineering and Methodology (TOSEM).*

EG Santana Jr, Gabriel Benjamin, **Jina Chun**, Melissa Araujo, Harrison Santos, David Freitas, Eduardo Almeida, Paulo Anselmo da Neto, Jiawei Li, Iftekhar Ahmed (2025). Which Prompt Works Best? An Empirical Evaluation of Prompting Techniques For Software Engineering Tasks. (*Arxiv*)

SKILLS

Research Skills
Natural Language Processing, Large Language Models, Multi-Agent Systems, Agentic Memory Systems, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Context Engineering, Developer Productivity Tools, Software Development Lifecycle (SDLC), Fine-tuning, Software Testing (JUnit, PyTest), Prompt Engineering, Model Training, Machine Learning, Dataset Curation and Labeling, Literature Review, Empirical Research, Qualitative and Statistical Analysis, Human Subjects Research

Programming Languages and Frameworks
Java, Python, JavaScript, TypeScript, Node.js, HTML, CSS, Django, React, Bootstrap

Machine Learning and AI Frameworks
PyTorch, TensorFlow, LangChain, scikit-learn, Hugging Face Transformers, OpenAI API

Data Analysis
Pandas, NumPy, Matplotlib, Seaborn, SQL, Data Visualization

Developer Tools and Platforms
Git, GitHub, Docker, Heroku, Linux, VS Code, Jupyter Notebook, Conda, Bash

PROJECTS

AI-Driven Mental Health Counseling

- Investigated whether open-source LLMs can replicate human counselor linguistic patterns.
- Applied various Natural Language Processing techniques to compare AI- and human-generated counseling responses.

Culture Explorer: Responsive Web Platform for Cultural Discovery

- Developed a responsive website showcasing global destinations and cultural experiences.

- Designed and deployed interactive gallery and destination pages optimized for responsiveness, usability, and cross-browser performance.

DineLog: Interactive Food Review and Sharing Web Platform

- Built a full-stack Django application supporting user registration, page creation, quote posting, and media uploads.
- Implemented modular apps with authentication, CRUD functionality, and Heroku-ready configuration. (Demonstration Video)

PROFESSIONAL SERVICE

Reviewer, IEEE Transactions on Reliability *April 2025*
Reviewer, ACM Transactions on Software Engineering and Methodology (TOSEM) *April 2024*
Reviewer, ACM Empirical Software Engineering *July 2025*

AWARDS AND HONORS

Chair's Award *Sept 2023*
Donald Bren School of Information and Computer Sciences, University of California, Irvine
Received upon admission to the Ph.D. program and granted in recognition of exceptional potential to make significant research contributions to the Informatics community.

Cum Laude *June 2021*
Boston University, College of Arts & Sciences
Awarded upon graduation in recognition of consistent academic excellence and distinguished achievement throughout undergraduate study.

Dean's List *August 2018- June 2021*
Boston University, College of Arts & Sciences
Received each semester throughout four years of study in recognition of consistent academic excellence and outstanding scholastic achievement.

STUDENT MENTOR

Mentor, ICS Honors Program *July 2023 - Ongoing*
University of California, Irvine

Mentor, Graduate International Connection Program *March 2023 - Ongoing*
University of California, Irvine

International Peer Mentor Program *Aug 2017 - June 2021*
Boston University